# Shiv Nadar University Chennai

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|---|---|---|
| Degree & Branch | B.Tech Artificial Intelligence & Data Science | Semester V |
| Subject Code & Name | CS3807 – Deep Learning Laboratory | AY: 2026–27 |

# Experiment 1
**Implementation of a Single Layer Perceptron for Binary Classification**

## 1. Objective

The objective of this experiment is to understand the concept of an artificial neuron and implement a Single Layer Perceptron from scratch for binary classification. Students will learn the perceptron learning algorithm, understand the importance of activation functions, visualize the learning process, and evaluate the classifier using a real-world dataset.

## 2. Background Theory

An Artificial Neuron is the fundamental building block of a neural network. It receives one or more input features, multiplies them by corresponding weights, adds a bias term, and applies an activation function to determine the output.

The Single Layer Perceptron is the earliest supervised learning algorithm capable of solving linearly separable binary classification problems.

### Mathematical Formulation

Weighted Sum

$$z = \mathbf{w}^T\mathbf{x} + b$$

where

- $\mathbf{x}$ : Input vector
- $\mathbf{w}$ : Weight vector
- $b$ : Bias
- $z$ : Linear combination

Prediction

$$\hat{y} = f(z)$$

Weight Update Rule

$$\mathbf{w}_{new} = \mathbf{w}_{old} + \eta(y - \hat{y})\mathbf{x}$$

Bias Update

$$b_{new} = b_{old} + \eta(y - \hat{y})$$

where $\eta$ denotes the learning rate.

## 3. Activation Functions

An activation function determines the output of a neuron based on the weighted sum of its inputs. It introduces non-linearity, enabling neural networks to learn complex patterns. Although the perceptron uses only the Step Activation Function, modern deep learning models employ differentiable activation functions for efficient optimization through backpropagation.

### Step Activation Function

$$f(z) = \begin{cases} 1, & z \geq 0 \\ 0, & z < 0 \end{cases}$$

The Step Activation Function produces a binary output and is therefore suitable only for linearly separable binary classification problems.

| **Activation** | **Output Range** | **Typical Applications** |
|---|---|---|
| Step | $\{0, 1\}$ | Classical Perceptron |
| Sigmoid | $(0, 1)$ | Binary Classification Output Layer |
| Tanh | $(-1, 1)$ | Hidden Layers (Traditional Networks) |
| ReLU | $[0, \infty)$ | Hidden Layers in CNNs and MLPs |
| Leaky ReLU | $(-\infty, \infty)$ | Avoiding Dying ReLU |
| Softmax | $(0, 1)$ | Multi-class Classification Output Layer |

**Note:** This experiment uses only the Step Activation Function. The remaining activation functions will be studied in subsequent experiments involving Multilayer Perceptrons and Deep Neural Networks.

## 4. Dataset Description

**Dataset:** Banknote Authentication Dataset
**Source:** UCI Machine Learning Repository
**Download Link:**
`https://archive.ics.uci.edu/dataset/267/banknote+authentication`

| | |
|---|---|
| Instances | 1372 |
| Features | 4 Numerical Features |
| Classes | 2 |
| Missing Values | None |
| Task | Binary Classification |

**Feature Description**

- Variance
- Skewness
- Curtosis
- Entropy

Target Class

- 0 – Authentic Banknote
- 1 – Forged Banknote

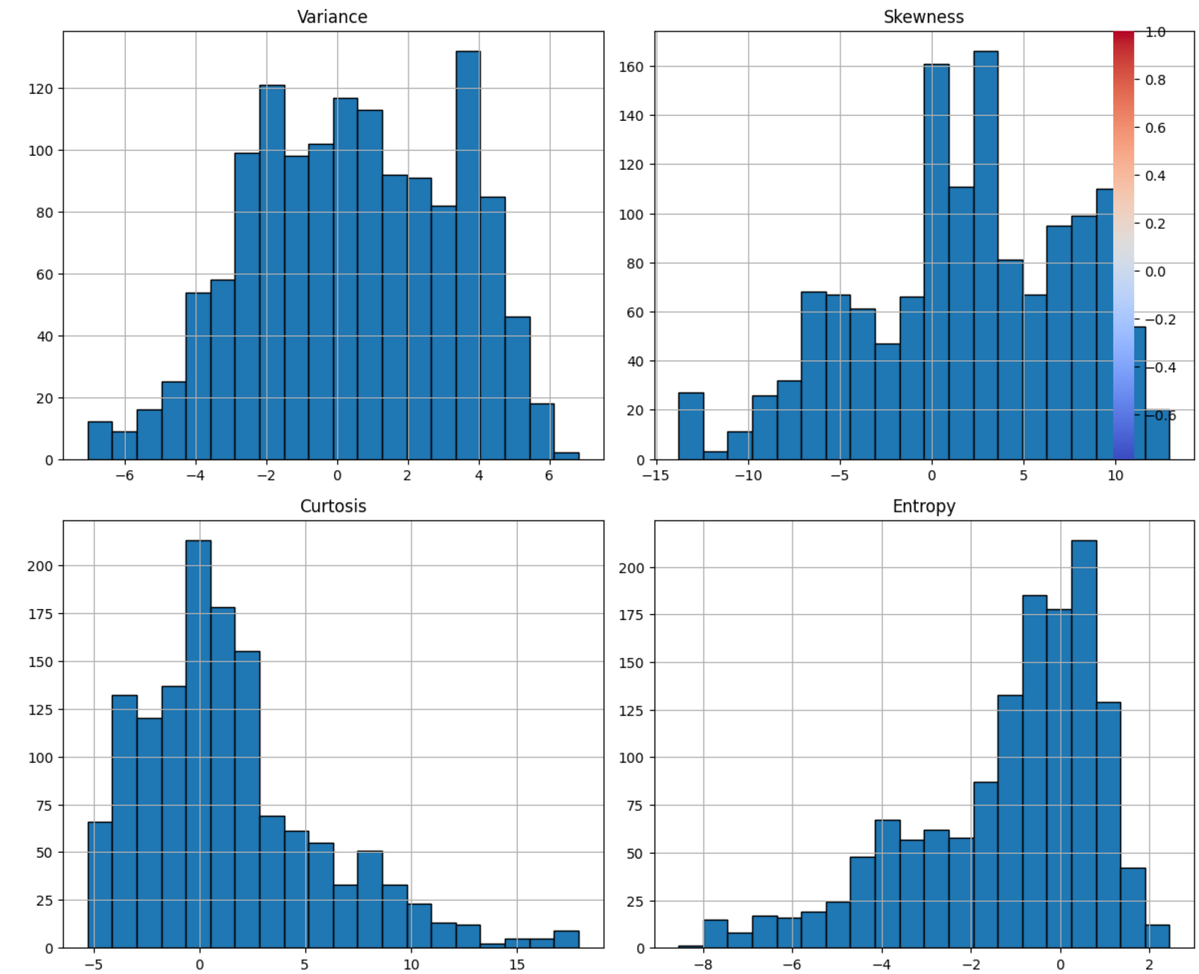
## 5. Source Code

The complete source code for this experiment, including dataset loading, exploratory data analysis, the from-scratch Single Layer Perceptron implementation, training, evaluation, and all plot generation, is available at the following GitHub repository:

`https://github.com/your-username/your-repository`

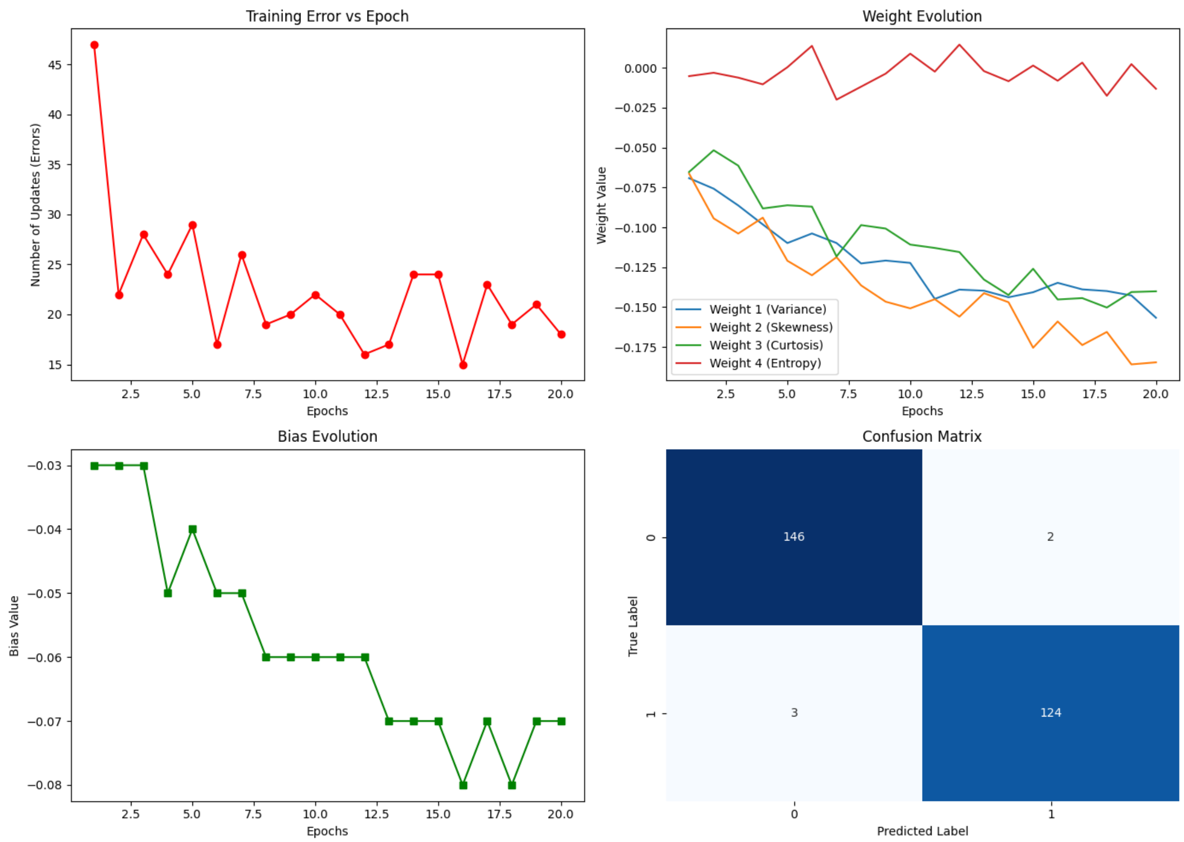
# 6. Mandatory Plots

### Feature Histograms, Correlation Heatmap, Scatter Plot and Boxplots



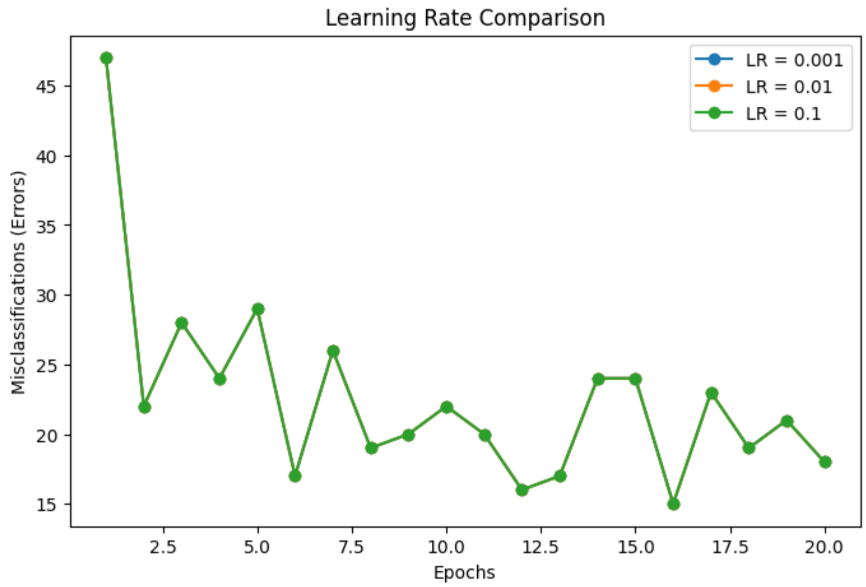

The histograms show that Variance and Entropy are roughly symmetric while Skewness and Curtosis are noticeably right-skewed, with Curtosis showing the most extreme outliers on the higher end. The correlation heatmap indicates a strong negative correlation between Variance and Class, and a moderate negative correlation between Skewness and Curtosis, while the scatter plot of Variance vs Skewness shows the two classes are largely, though not perfectly, linearly separable. The boxplots further confirm that Skewness and Curtosis have the widest spread and the most outliers among the four features.

### Training Error vs Epoch, Weight Evolution, Bias Evolution and Confusion Matrix



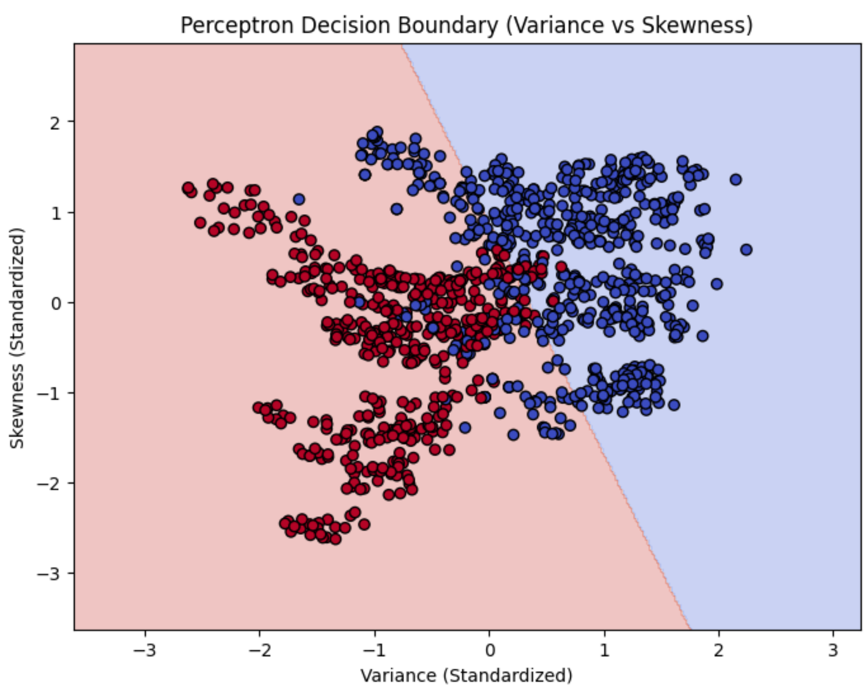

The Training Error vs Epoch plot shows a sharp drop in misclassifications from 47 to 22 within the first two epochs, after which the error oscillates between roughly 15 and 29 without fully converging to zero, indicating the classes are not perfectly linearly separable. The Weight Evolution plot shows that the weights for Variance and Skewness decrease steadily and stabilize around $-0.15$ to $-0.18$, while the Bias Evolution plot shows the bias drifting downward in discrete steps from $-0.03$ to $-0.07$ as training progresses. The Confusion Matrix shows only 2 false positives and 3 false negatives out of 275 test samples, confirming the strong classification performance reported in the evaluation metrics.

## Learning Rate Comparison




The misclassification curves for learning rates 0.001, 0.01 and 0.1 overlap almost exactly, since the sign of each weight update (and hence the prediction at every step) is identical for zero-initialized weights and only the magnitude of the weights scales with $\eta$. This shows that for this dataset the perceptron's convergence behaviour (number of epochs to stabilize, oscillation pattern) is essentially independent of the learning rate, and only the final weight magnitudes differ.

## Decision Boundary (Optional)



Using only Variance and Skewness, the perceptron learns a linear decision boundary that separates most authentic (blue) banknotes from forged (red) banknotes, correctly capturing the general trend that higher Variance corresponds to authentic notes. A visible band of misclassified points near the boundary confirms that these two features alone are not perfectly linearly separable, which is why the training error in the earlier plot never converges to zero.

## 7. Performance Tables

**Training Summary**

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|---|---|
| Dataset Size | 1372 samples |
| Train/Test Split | 1097 / 275 (80% / 20%) |
| Learning Rate | 0.01 |
| Epochs | 20 |
| Final Weights | [-0.1567, -0.1846, -0.1401, -0.0132] |
| Final Bias | -0.0700 |
| Accuracy | 0.9818 |
| Precision | 0.9841 |
| Recall | 0.9764 |
| F1-score | 0.9802 |

**Epoch-wise Learning** (Variance and Skewness weights shown; $\eta = 0.01$)

| Epoch | Errors | Weight 1 (Variance) | Weight 2 (Skewness) | Bias |
|---|---|---|---|---|
| 1 | 47 | -0.0692 | -0.0663 | -0.0300 |
| 2 | 22 | -0.0758 | -0.0944 | -0.0300 |
| 3 | 28 | -0.0863 | -0.1039 | -0.0300 |
| 4 | 24 | -0.0982 | -0.0939 | -0.0500 |
| 5 | 29 | -0.1098 | -0.1210 | -0.0400 |
| ⋮ | ⋮ | ⋮ | ⋮ | ⋮ |
| 16 | 15 | -0.1348 | -0.1590 | -0.0800 |
| 17 | 23 | -0.1389 | -0.1738 | -0.0700 |
| 18 | 19 | -0.1399 | -0.1656 | -0.0800 |
| 19 | 21 | -0.1427 | -0.1859 | -0.0700 |
| 20 | 18 | -0.1567 | -0.1846 | -0.0700 |

## 8. Discussion and Conclusion

The Single Layer Perceptron trained from scratch on the Banknote Authentication dataset achieved a strong test accuracy of 98.18%, with precision, recall and F1-score all above 0.97, confirming that the four features (Variance, Skewness, Curtosis and Entropy) make the two classes almost, though not perfectly, linearly separable. The training error did not converge to exactly zero within 20 epochs, oscillating between 15 and 29 misclassifications per epoch after an initial sharp drop, which indicates a small number of samples lie on the wrong side of every possible linear boundary. The learning rate comparison showed that, for this zero-initialized perceptron, the number of misclassifications per epoch is essentially unaffected by $\eta \in \{0.001, 0.01, 0.1\}$, since the learning rate only scales the magnitude of the weight updates and not their sign. When restricted to only two features (Variance and Skewness) for the decision boundary visualization, the model still performed reasonably but with visibly more overlap near the boundary, showing that Curtosis and Entropy do contribute useful separating information. The custom implementation's accuracy (98.18%) was very close to that of Scikit-learn's built-in Perceptron (98.55%), validating the correctness of the from-scratch implementation of the weighted sum, step activation, and the perceptron learning rule. Overall, the experiment demonstrates both the strengths of the perceptron as a simple, interpretable linear classifier on well-behaved data, and its fundamental limitation on data that is not perfectly linearly separable.

## 9. References


1. F. Rosenblatt, "The Perceptron," *Psychological Review*, 1958.
2. I. Goodfellow, Y. Bengio and A. Courville, *Deep Learning*, MIT Press, 2016.
3. C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
4. S. Haykin, *Neural Networks and Learning Machines*, Pearson, 2009.
5. UCI Machine Learning Repository – Banknote Authentication Dataset.
6. Scikit-learn Documentation: Perceptron.